

Day 2 Worksheet: Anatomy of Roots and Stems

Dicot Root vs Monocot Root | Dicot Stem vs Monocot Stem | Stele & Casparian Strips

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Student Name: Date: Time Allowed: 30 Mins Max Marks: 20 Marks Obtained:

Day 2 Focus Review: • Root Endodermis has *Casparian strips* (suberin). Dicot root = 2–4 xylem patches & small pith; Monocot root = Polyarch (>6 patches) & large pith. • Dicot Stem = Collenchymatous hypodermis, ring of open vascular bundles, starch sheath, large pith. • Monocot Stem = Sclerenchymatous hypodermis, scattered closed vascular bundles, bundle sheath, water cavities, no pith.

SECTION A: FILL IN THE BLANKS & OBJECTIVE (1 MARK EACH)

1. The Casparian strips are deposition of water-impermeable waxy material called

1M

2. In dicot root, the number of xylem and phloem patches is usually:

1M

(a) 2 to 4 (b) More than 6 (c) 10 to 15 (d) Single only

3. The innermost layer of the cortex in dicot stem is rich in starch grains and is termed as:

1M

(a) Casparian strip (b) Starch sheath (c) Pericycle (d) Epiblema

4. Sclerenchymatous hypodermis and scattered vascular bundles are diagnostic of:

1M

(a) Dicot Stem (b) Monocot Stem (c) Dicot Root (d) Monocot Root

SECTION B: SHORT ANSWER & REASONING QUESTIONS (2 MARKS EACH)

5. Define Stele. List the three constituent tissues that form the stele.

2M

6. What is Conjunctive tissue? What is its role during secondary growth in dicot roots?

2M

7. Why do monocotyledonous roots not undergo any secondary growth?

2M

SECTION C: HIGH-YIELD BOARD EXAM DIFFERENCES (3 & 4 MARKS)

8. State any three major anatomical differences between Dicot Root and Monocot Root.

3M

9. Complete the comparative table between Dicot Stem and Monocot Stem:

4M

Feature	Dicot Stem	Monocot Stem
Hypodermis Nature		
Vascular Bundle Arrangement		
Cambium (Open / Closed)		
Presence of Pith		